

Incorporate electrical impedance tomography sensors in venous filter to detect increased pulsatility as an early indicator of flow obstruction

Executive Technology Brief

Domain: ml_data

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'obstruction_of_flow' for venous filter: 9 MAUDE reports across 8 distinct devices (20161117 to 20260505); root-cause evidence from 0 recall records and 9 event narratives (custody attached in the campaign report). MAUDE limitations: incidence UNKNOWN, causation UNVERIFIED (Art. XXI.5). Existing solutions fail because: Why existing solutions fail, per retrieved evidence: 9 MAUDE event narratives from this mechanism cluster (mdr_text, custody attached). Statements are the providers' own words, custody-attached; the aggregation across records is AI_INFERENCE (bounded by the re Unresolved gap: Across the 7 relevant retrieved attempts (a bounded, non-exhaustive set), none measurably addresses: conditions, devices, events, failure, function, injury, intervention, introducing, maintain, matching, modes, must, problem, producing, recurring, reported, un

Mechanism: Regional pulsatility amplitude increases with downstream flow obstruction

Architecture: PROPOSED architecture (no physical system exists): data ingestion and labeling pipeline; feature / representation layer; model + uncertainty quantification; deployment target + monitoring loop — realizing: Incorporate electrical impedance tomography sensors in venous filter to detect increased pulsatility as an early indicator of flow obstruction

Governing Model: Data architecture / machine learning governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

ML-002: Bayes' theorem (posterior inference). Expression: $P(y|x) = P(x|y) * P(y) / P(x)$. Applicability to this invention: CONDITIONAL; holds under: probability model correctly specified; priors explicit. Assumptions: priors and likelihoods are MODEL_DERIVED until validated on deployment data. Engaged variables: NOT ESTABLISHED.

ML-003: Empirical risk. Expression: $R_{\text{emp}} = \text{sum}(\text{loss}(f(x_i), y_i)) / n$. Applicability to this invention: CONDITIONAL; holds under: samples independent and representative of the deployment distribution. Assumptions: empirical risk on a non-representative sample says nothing about deployment risk (distribution shift). Engaged variables: NOT ESTABLISHED.

ML-001: Expected utility of a decision rule. Expression: $EU = \text{sum}_y P(y | x) * U(a(x), y)$. Applicability to this invention: CONDITIONAL; holds under: utility table fully specified; probabilities calibrated. Assumptions: utility weights are BUYER_DEFINED/MODEL_DERIVED, never measured. Engaged variables: NOT ESTABLISHED.

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.44573539382492045, EIG-per-cost 0.44573539382492045; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.